

Intermediate Quantitative Methods

The logic of statistical inference

Kevin Munger

Week 1

European University Institute, SPS

Today

- From sample to population

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- Standard errors and intervals

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- Hypothesis tests and p-values

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- Standard errors and intervals
- Hypothesis tests and p-values
- Power, and what low power does

From sample to population

The problem

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- We care about a population

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- We observe a sample
- How far can one sample take us?

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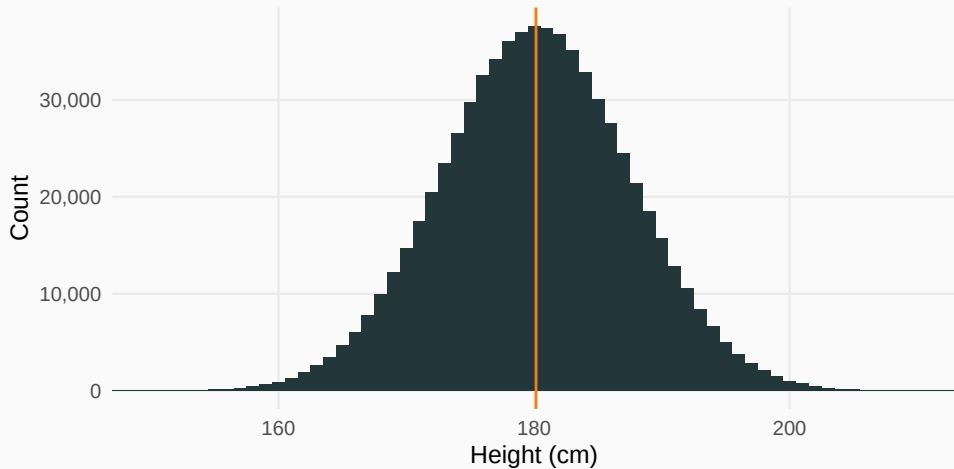
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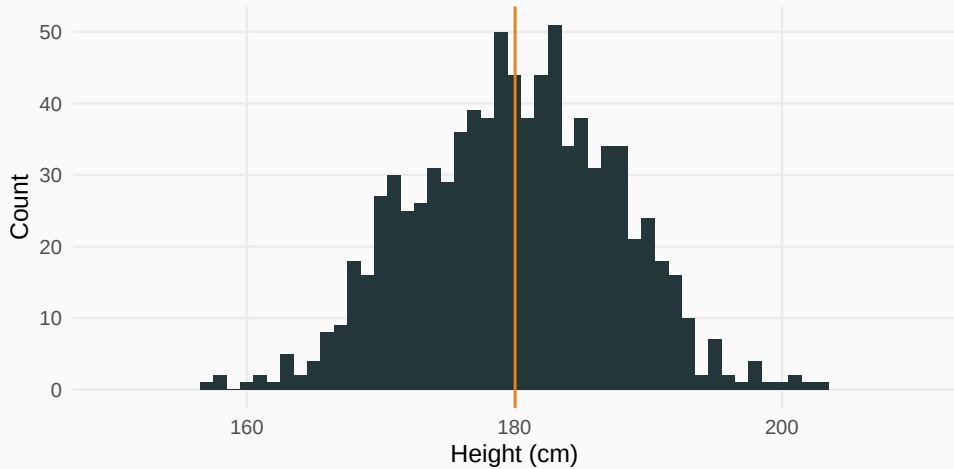
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- Notation: Greek letters (μ, σ) for the population, Latin letters ($\bar{x}, s$) for the sample
- Defining the estimand is often the hard part: week 3

Population: 700,000 simulated heights



One random sample, $n = 859$



Three distributions

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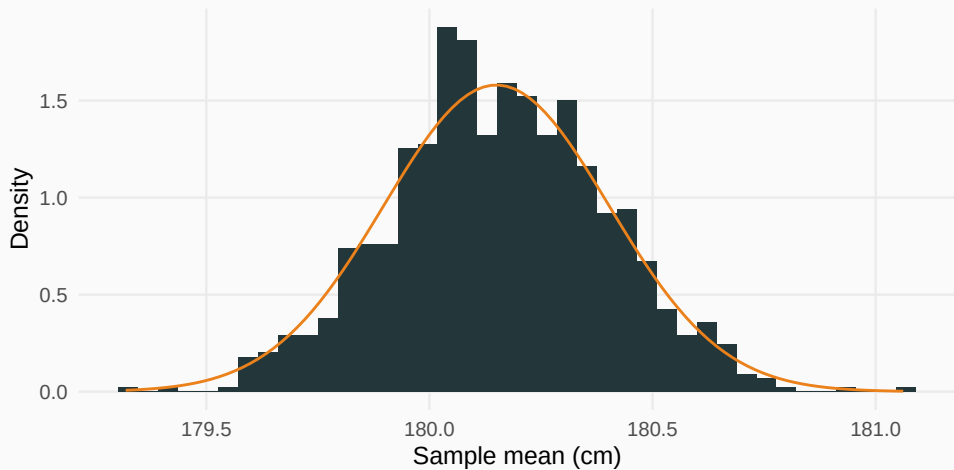
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- Sampling distribution: the distribution of $\bar{x}$ across repeated samples
- Never observed, only derived from theory or simulated

Means of 1,000 samples, $n = 859$



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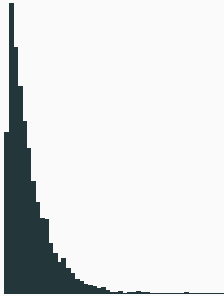
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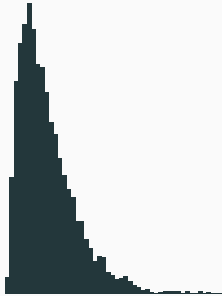
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- Here: $\sigma = 7.4$, $n = 859$, so $SE \approx 0.25$ cm

Skewed population, means of 5,000 samples

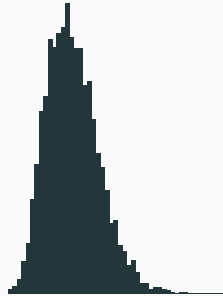
Population



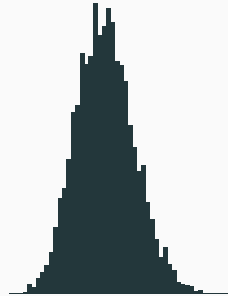
Means, $n = 2$



Means, $n = 10$



Means, $n = 50$



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- Why normal-based tests work far beyond normal data

Standard errors and intervals

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- Easy to confuse, answer different questions

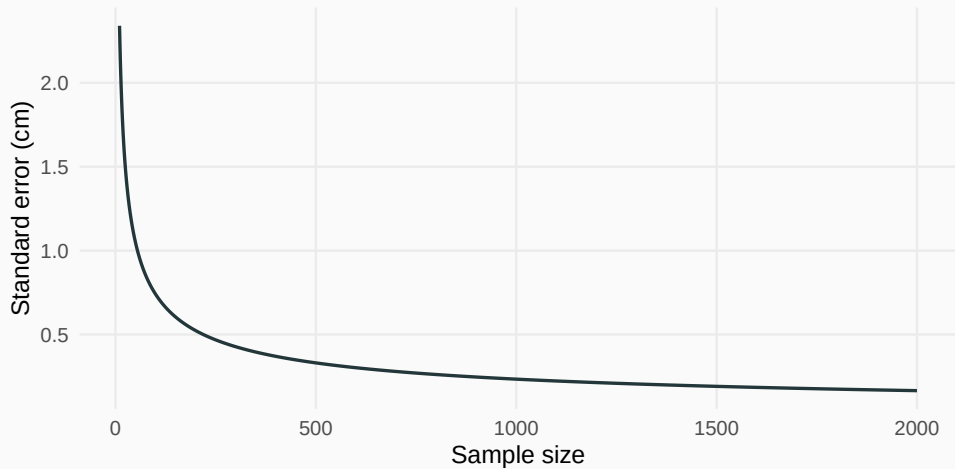
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- σ unknown, so we plug in s : $\widehat{SE} = s / \sqrt{n}$

Standard error by sample size, $\sigma = 7.4$



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- 68% of the mass within ± 1 , 95% within ± 1.96 , 99% within ± 2.58

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- Depends on degrees of freedom ($n - 1$): 2.09 at $n = 20$, 1.98 at $n = 100$, then ≈ 1.96

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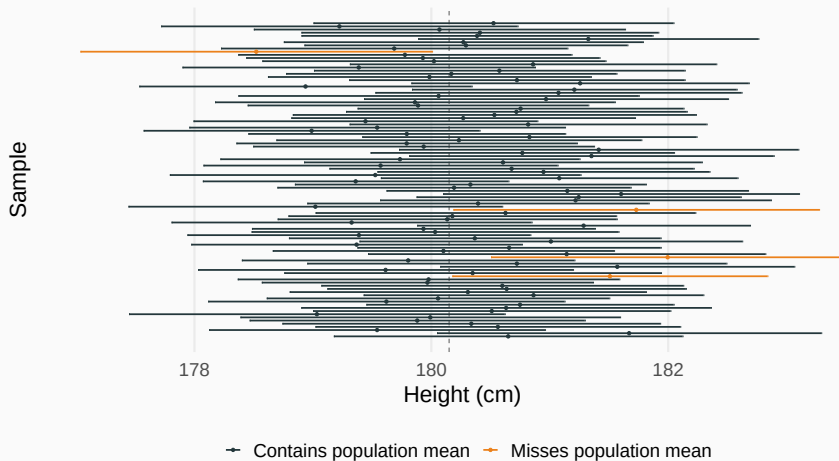
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- 95% CI: $180.05 \pm 1.96 \times 0.26 = [179.55, 180.56]$
- $\mu = 180.15$: inside, this time

100 samples of 100, 100 95% intervals



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- Not a 95% probability that μ is in this interval

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- Reject H_0 if $p < \alpha$, by convention $\alpha = 0.05$

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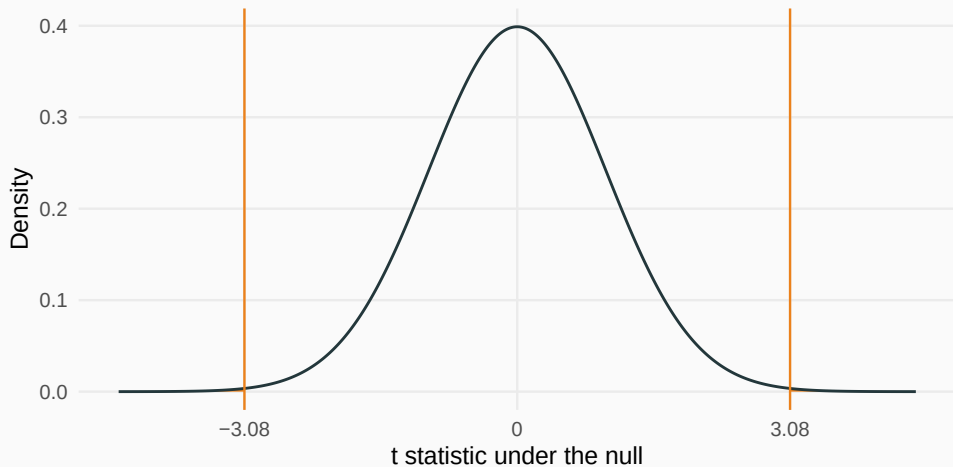
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- Estimated difference 2.44 years, $\widehat{SE} = 0.79$
- $t = 2.44/0.79 \approx 3.08$
- $p = 0.002$: about 2 in 1,000 samples under H_0

Null distribution, $t = \pm 3.08$



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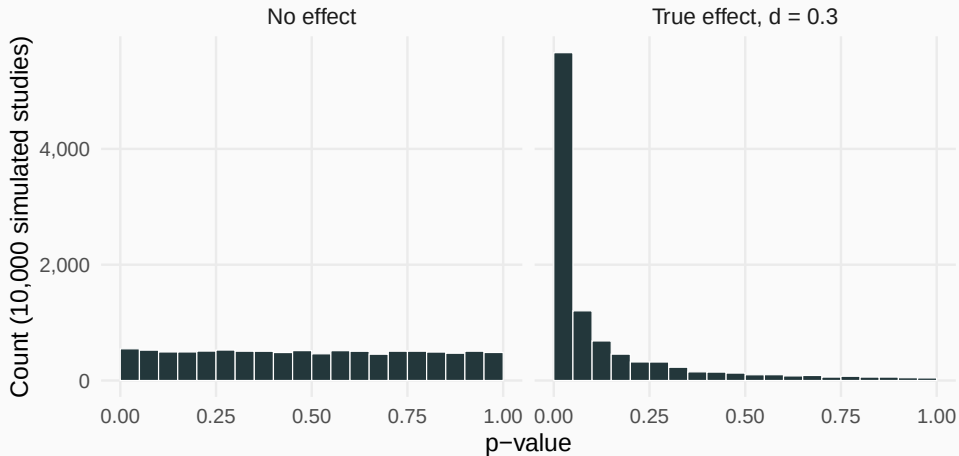
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- Not $P(H_0 \text{ true})$
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- Not the probability of replication
- Under H_0 , every p-value between 0 and 1 is equally likely

p-values from 10,000 simulated studies, $n = 100$ per group



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- Power = $1 - \beta$: probability of detecting an effect that is there

Power

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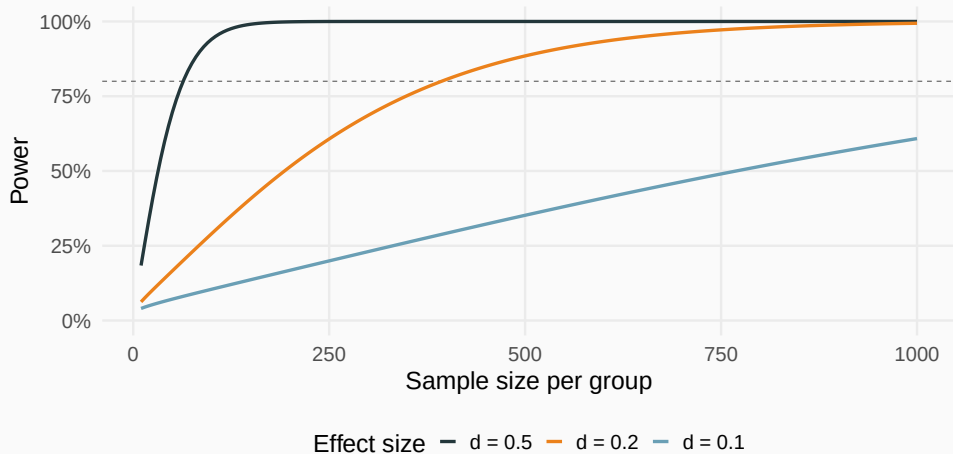
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- Usually: d , $\alpha = 0.05$ and 80% power give the n you need

Power by sample size, two groups, $\alpha = 0.05$



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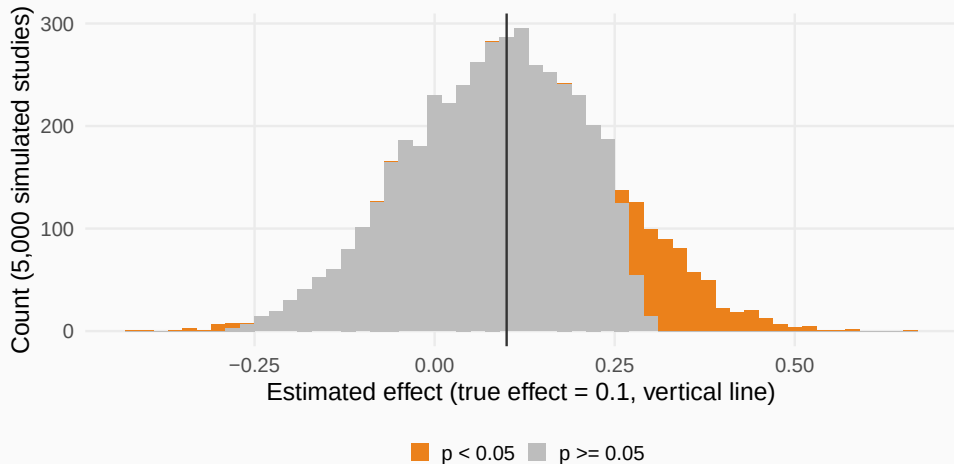
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- Arel-Bundock et al. (2026)

5,000 studies, true effect 0.1, $n = 100$ per group



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- Type M (magnitude) error: significant estimates exaggerate the true effect
- Type S (sign) error: significant estimates with the wrong sign
- Real problem is the combination with publication bias: a higher percentage of positive findings are false positives

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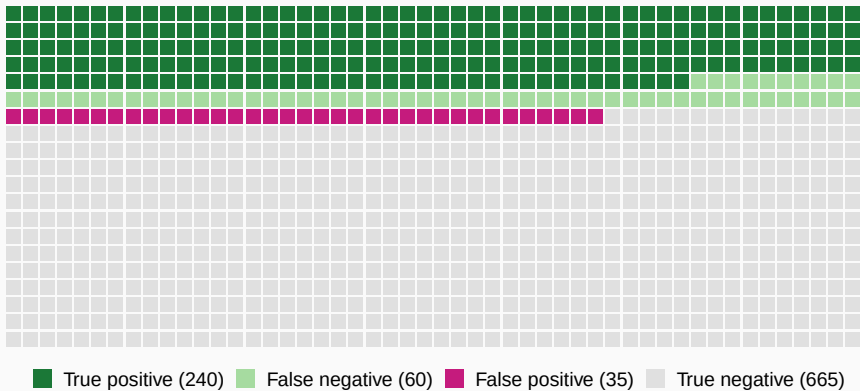
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- Depends on power
- Depends on the share of tested hypotheses that are true

1,000 hypotheses: 30% true, 80% power, $\alpha = 0.05$



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- More false positives than true ones